Crop Rotation and Agricultural Diversification

Agriculture RAG Knowledge Base • Educational reference document

Crop rotation means growing different crops in a planned sequence over multiple seasons. Diversification can also include intercropping, cover crops, and integration of crops with livestock where appropriate.

Rotation can improve agricultural resilience because different crops have different nutrient demands, root systems, and pest associations. Including legumes in a rotation can contribute biologically fixed nitrogen to the farming system, although the amount available to a following crop depends on biomass, management, decomposition, and other conditions.

Rotations can interrupt the life cycles of some pests and diseases. They can also influence weed pressure because changing planting dates and crop competition can alter which weeds succeed. Cover crops can protect soil from erosion and contribute organic residues.

A rotation should be designed around local climate, soil, market conditions, available labor, water, equipment, and crop suitability. A theoretically good rotation may not be practical if a farmer cannot sell the crop or lacks the equipment needed to establish it.

Diversification is therefore both an agronomic and economic decision. Farmers can compare expected resource requirements, risks, market opportunities, and effects on soil and nutrient management before choosing a rotation plan.

Reference basis

• FAO, Integrated Plant Nutrient Management

• FAO, soil and conservation agriculture resources

Note: This document is a project knowledge-base document prepared for a student RAG demonstration. It is not a substitute for crop-specific agronomic advice, soil testing, product labels, or local agricultural recommendations.